ABSTRACT OF THE DISCLOSURE

A method for dynamically composing and provisioning a logical, bare-metal computer from physically disaggregated pools of hardware resources. The method comprises: a control plane receiving a request specifying a quantity of computer resources (e.g., number of CPUs, amount of memory); identifying available, disaggregated resources from their respective pools; transmitting a command to a data fabric switch to establish a set of persistent optical links interconnecting the identified resources; and exposing the resulting collection of interconnected resources as a single, unified, bare-metal computer to a user or task. Upon task completion, the links are dissolved, and the resources are returned to the pools.